

CircuitMedic

Firmware debugging
with traceable evidence

Arduino Uno + HC-SR04 prototype

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The debugging gap

Firmware

```
delayMicroseconds(4);
```

A short trigger delay appears in the sensor routine.

Technical documentation

At least 10 μ s HIGH

The HC-SR04 datasheet specifies the trigger input.

The developer must connect the code to the requirement.

First users and their task

Engineering students

Learning how firmware behavior relates to sensor specifications.

Student robotics teams

Reviewing sensor routines before integrating them into a larger robot project.

User task

Locate a supported issue, inspect its evidence, apply a change and check the firmware again.

Evidence inside the review workflow

Hardware context

Board

Arduino Uno

Supported component

HC-SR04

TRIG pin symbol

trigPin

ECHO pin symbol

echoPin

Demo examples

Load broken example

Load fixed example

Analysis complete — 3 issues

Analysis complete

Location

Direct rule match

broken_robot.ino:12

Problematic code line

```
delayMicroseconds(4); // Bug: needs at least 10 us
```

Observed condition

```
delayMicroseconds(4)
```

Documented requirement

The IIC-SR04 requires the TRIG input to be held HIGH for at least 10 microseconds.

Why this is a mismatch

The observed trigger delay is shorter than the documented minimum.

Suggested fix

Hold TRIG HIGH for at least 10 microseconds, for example `delayMicroseconds(10)`.

Document evidence

Tech Support: services@elecfreaks.com Ultrasonic Ranging Module HC - SR04 Product features: JRC RFR Ultrasonic ranging module HC - SR04 provides 2cm - 400cm non-contact measurement function, the ranging accuracy can reach to 3mm. The module includes ultrasonic transmitters, receiver and control circuit. The basic principle of work: (1) Using IO trigger for at least 10us high level signal, (2) The Module automatically sends eight 40 kHz and detect whether there is a pulse signal back. (3) If the signal back, through high level, time of high output IO duration is the time from sending ultrasonic to returning.

Source file: [sr04_original.pdf](#) - page 1 - chunk file: [sr04_original.p1.c1](#) - retrieval similarity 0.580

Open source document

Retrieval similarity measures how closely the document passage matches the search query. It is not the probability that the diagnosis is correct.

> HIGH - Echo measurement relies on the default pulsein timeout

> MFDIUM - No echo result is used as a valid distance

© AI Debugging Explanation

Inspect

Code location
Requirement
Suggested fix
Source passage

Firmware upload, in-browser editing and JSON report export

A fix the user can verify

03

Supported findings
in the broken
example

02

After changing the
trigger delay to 10 μ s

00

In the corrected
example

The user edits the code and runs the checks again.

The corrected example also adds an explicit timeout and handles no echo.
Zero findings covers the supported checks only.

System architecture and the role of AI

- | | | |
|-----------|--------------------------------|---|
| 01 | Firmware analysis | Deterministic checks identify supported code patterns. |
| 02 | Document retrieval | Local sentence embeddings retrieve relevant passages. |
| 03 | Diagnostic report | Code findings are paired with source metadata and fixes. |
| 04 | Optional AI explanation | An OpenAI model explains existing findings and uncertainty. |

Unknown evidence IDs are rejected. The base report survives an AI failure.

Regression checks for known failure cases

43

automated tests passed

15 existing tests

28 new regression tests

Scope: Arduino Uno + HC-SR04. Full C++ analysis and physical hardware validation remain outside this prototype.

Review cases covered

Multiline calls and same-line guards
Conditional returns and later uses
Millisecond and immutable delays
Source-like strings and comments

User value and the next validation step

Available in the prototype

A traceable explanation beside each supported finding.

A repeatable edit-and-check workflow.

To measure with users

Time to identify the issue.

Ability to explain the fix.

False alarms and missed issues on independent firmware.

No user adoption or time-saving results are claimed yet.

Next steps

Validate

Independent
firmware
and student
feedback

Improve

Refine supported
checks
using observed
failures

Extend

Additional
components
after validation

CircuitMedic makes firmware findings inspectable.

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github.com/41iclalsevdeyavuz-crypto/CircuitMedic